

Tawit Uthaicharoenpong

LinkedIn: www.linkedin.com/in/tawit-u • **Mobile:** +66 815662735 • **Email:** iamtawit@gmail.com

GitHub: github.com/tawituthai/showcases

Base location: Bangkok, Thailand • **Citizenship:** Thai

PROFESSIONAL SUMMARY

Principal Electrical Engineer with 15+ years leading the full hardware lifecycle for autonomous vehicle fleets — schematic capture, PCB layout, motor and actuator control, embedded firmware, and high-reliability manufacturing (MTTF, DFM, IPC Class 3). Building a documented hardware portfolio (power distribution, distributed avionics, control-authority systems) that independently converges on patterns used in spacecraft design — see github.com/tawituthai/showcases — alongside a Master's in Space Systems Engineering at Johns Hopkins (in progress).

CORE COMPETENCIES

- Robotics & Motor Control: Drive-by-Wire actuation systems, closed-loop real-time control
- Embedded Systems: C/C++, Python, low-level firmware, real-time control loops, fault detection
- Hardware Design: Schematic capture, PCB layout (high-speed/mixed signal), power electronics, cable harness design — Altium Designer, KiCAD
- Reliability & Manufacturing: MTTF analysis, thermal profiling, DFM/DFA, IPC Class 3, vendor management, BOM optimization, NPI
- Space Systems: ADCS fundamentals, radiation-hardened electronics, N+R redundant power architectures, dual-redundant CAN bus
- Tools: Altium Designer, KiCAD, SolidWorks, Fusion 360, Python

PROFESSIONAL EXPERIENCE

Venti Technologies
Principal Electrical Engineer

Singapore (Remote from Bangkok)
August 2022 - Present

- Led end-to-end development of Drive-by-Wire actuation control boards — schematic capture through PCB layout, fabrication, and validation — resolving stack-up and impedance issues pre-production.
- Engineered safety-critical Power Distribution Units with calculated MTTF and thermal profiling, optimizing component derating to extend safe operating temperature range.
- Developed embedded C firmware for steering, throttle, and braking actuation control under hard real-time constraints, with fault detection built into the control loop.
- Championed a modular hardware platform strategy adaptable across multiple autonomous vehicle programs, reducing design cycle time and inventory complexity.
- Led component-level debugging and root-cause failure analysis of prototype boards, feeding findings back into design.

Motional Inc
Software Engineer

Singapore
April 2021 – August 2022

- Led hardware integration for the Remote Vehicle Assistance system, enabling human-in-the-loop trajectory control for autonomous vehicles in edge-case scenarios.
- Migrated the legacy SimpleComm communication interface to a modern IDL-based standard across the autonomous vehicle software stack.
- Implemented C++ path-planning algorithms with the Planning and Control team, improving vehicle agility in dense traffic environments.

nuTonomy (an APTIV company)
Principal Electrical Engineer

Singapore
August 2015 - April 2021

- Designed electronic circuits for computer-controlled vehicle actuation (steering, throttle, braking) with manual override, and led power-provisioning design mitigating in-rush current for remote software deployment to autonomous vehicle compute systems.
- Oversaw fabrication and assembly of safety-critical automotive hardware to IPC Class 3 standards; developed validation test plans ensuring compliance with automotive safety requirements.
- Managed physical integration of compute units, sensors, and power systems into autonomous test vehicles (Renault ZOE, Peugeot 3007), establishing repeatable installation procedures for fleet expansion.
- Led cybersecurity hardening of the power control unit against unauthorized actuation.

SINGAPORE-MIT ALLIANCE FOR RESEARCH AND TECHNOLOGY (SMART)
Research Engineer

Singapore
August 2012 - July 2015

- Built and maintained general I/O devices and power circuits for experimental mobility-on-demand vehicles, transitioning breadboard prototypes to field-ready units.
- Created manufacturing documentation — assembly drawings and wiring diagrams — standardizing the build process across the vehicle integration team.

EDUCATION

Johns Hopkins University — MS, Space Systems Engineering
Coursework completed: EN.675.600 Systems Engineering for Space, EN.675.601 Fundamentals of Engineering Space Systems I

In Progress

UNIVERSITY OF NEW SOUTH WALES (UNSW)
Master of Engineering (Electrical Engineering)

Sydney, Australia
2009-2010

CHULALONGKORN UNIVERSITY
Bachelor of Engineering (Electrical Engineering)

Bangkok, Thailand
2002-2006

CERTIFICATION

IPC/WHMA-A-620
Requirements and Acceptance for Cable and Wire Harness Assemblies.

March 2019 - March 2021

UNITED STATES PATENTS

Co-inventor · Traffic Signal Control Method And Traffic Signal Controller
US 9,601,013 B2

[Filing] Inventor · Controlling Power Of Electronic Devices On A Vehicle · 17/017,656